# LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

**ANSWER KEY & MARKING SCHEME**

## Practice Pack: Work and Energy

*Chapter / Topic: Topic-based: use with whichever Class 9 book your students follow*

*For teacher use only. Do not distribute to students.*

### LEVEL 1: CONCEPTS

**Q1.** Define work and give its SI unit.

**Answer:** Work is done when a force moves an object in the direction of the force: W = F × s. The SI unit is the joule (J). 1 J = 1 N × 1 m.

**Q2.** Give two situations where a force acts but no work is done.

**Answer:** (1) Pushing a wall that does not move (no displacement). (2) Carrying a load on your head while walking on level ground: the force of gravity is perpendicular to the displacement, so gravity does no work.

**Q3.** Define power and give its SI unit.

**Answer:** Power is the rate of doing work (P = W/t). The SI unit is the watt (W) = 1 J/s.

**Q4.** State the law of conservation of energy.

**Answer:** Energy can neither be created nor destroyed. It can only change from one form to another, so the total energy stays the same.

**Q5.** How many joules are there in 1 kWh?

**Answer:** 1 kWh = 1000 W × 3600 s = $3.6 \times 10^6$ J.

### LEVEL 2: NUMERICALS

**Q6.** A force of 7 N moves an object 8 m in the direction of the force. How much work is done?

**Answer:** W = F × s = 7 × 8 = 56 J.

**Q7.** Find the kinetic energy of a 15 kg object moving at 4 m/s.

**Answer:** KE = ½$mv^2$ = ½ × 15 × $4^2$ = 120 J.

**Q8.** Find the potential energy of a 10 kg object at a height of 6 m. (g = 9.8 m/$s^2$)

**Answer:** PE = mgh = 10 × 9.8 × 6 = 588 J.

**Q9.** A lamp uses 1000 J of electrical energy in 10 s. What is its power?

**Answer:** P = W/t = 1000/10 = 100 W.

**Q10.** How much work is needed to increase the speed of a 1500 kg car from 30 km/h to 60 km/h?

**Answer:** u = 30/3.6 ≈ 8.33 m/s, v = 60/3.6 ≈ 16.67 m/s. W = change in KE = ½$m(v^2 - u^2)$ = ½ × 1500 × (277.8 – 69.4) ≈ $1.56 \times 10^5$ J (156,250 J).

**Q11.** A 1500 W heater runs for 10 hours. How much energy does it use, in kWh?

**Answer:** E = P × t = 1.5 kW × 10 h = 15 kWh.

### LEVEL 3: CHALLENGE (FOUNDATION)

**Q12.** Two girls, each weighing 400 N, climb a rope to a height of 8 m. Girl A takes 20 s and girl B takes 50 s. Find the power of each.

**Answer:** Work by each = 400 × 8 = 3200 J. Power of A = 3200/20 = 160 W. Power of B = 3200/50 = 64 W.

**Q13.** A boy of mass 50 kg runs up a staircase of 45 steps in 9 s. Each step is 15 cm high. Find his power. ($g = 10\ m/s^2$)

**Answer:** Height = 45 × 0.15 = 6.75 m. Work = mgh = 50 × 10 × 6.75 = 3375 J. Power = 3375/9 = 375 W.

**Q14.** An object of mass 40 kg is raised to a height of 5 m. Find its potential energy. If it is allowed to fall, find its kinetic energy when it is halfway…

**Answer:** PE = mgh = 40 × 10 × 5 = 2000 J. Halfway down, PE = 40 × 10 × 2.5 = 1000 J. By conservation of energy, KE = 2000 – 1000 = 1000 J.